

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457772.879 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.047
Transit depth (Rp/Rs)^2	2.64 +/- 1.53 [%]
Orbital Inclination (inc)	80.58 +/- 2.89
Airmass coefficient 1 (a1)	1.43 +/- 0.81
Airmass coefficient 2 (a2)	-0.23 +/- 0.43
Scatter in the residuals of the lightcurve fit is	56.13 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	4.75
Transit Duration (day)	0.047 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.163 $\pm$ 0.047 vs 0.1368 (deviation 0.46 σ)
Duration fit vs published	0.047 d vs 0.0758 d (deviation 0.72 σ)
Mid-time O-C	-5.5 min
Depth SNR	2.9
Residual scatter	56.13 %

Light curve

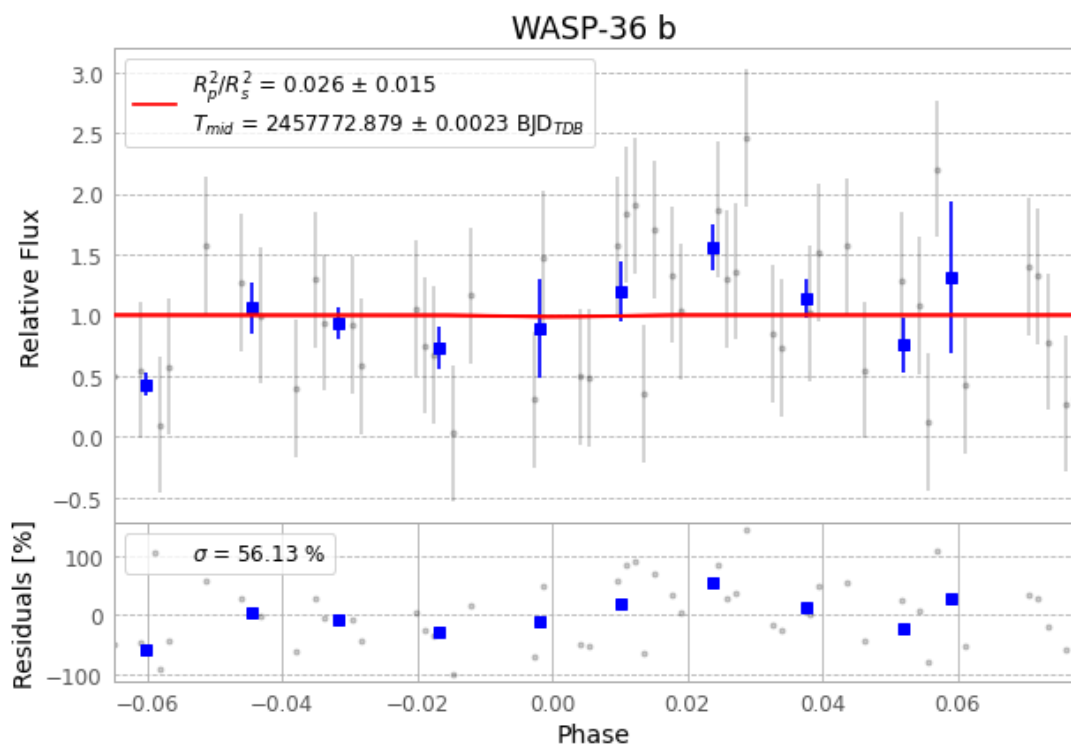


Figure 1 — FinalLightCurve_WASP-36 b_2017-01-18.png (SHA-256 75ad51dc65bca56f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [220, 444], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7d0d9099011847d4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

107 FITS frames (70 MB), WASP-36170119063312.FITS → WASP-36170119115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-170119.log (491cd21454c5...), FinalLightCurve_WASP-36 b_2017-01-18.png (75ad51dc65bc...), FinalLightCurve_WASP-36 b_2017-01-18.csv (7d81d6c893d0...)

Compute resources

Wall time 190.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 20:32Z.